

Agents allow special kinds of culmination (not just special non-culmination)

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TOM 18

May 22, 2026

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Aspect is our temporal perspective on events.

Perfective aspect tends to feature:

- Event as an undifferentiated whole
- Events one after another in time
- Events completely contained within reference time

E.g.:

Perfective: *When I came in, she laughed and took a picture.*

Imperfective: *When I came in, she was laughing and taking a picture.*

Perfectives sometimes entail that events *culminated*, that is, reached a particular specified endpoint.

E.g.:

Perfective: *The log floated to the sea.*

Imperfective: *The log was floating to the sea.*

Not all verb phrases have any concept of culmination. Those that do are **telic**.

E.g.:

The log floated to the sea.

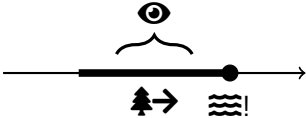
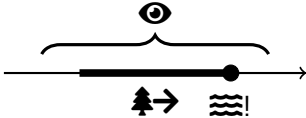
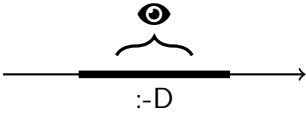
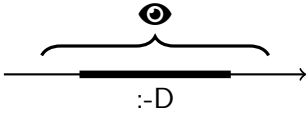
For an **atelic** verb phrase, perfective and imperfective aspects have similar entailments.

E.g.:

Perfective: *She laughed.*

Imperfective: *She was laughing.*

Naive Picture of Perfectivity and Telicity

	Imperfective	Perfective
Telic	<i>The log was floating to the sea.</i>  → No culmination	<i>The log floated to the sea.</i>  → Culmination
Atelic	<i>She was laughing.</i>  → No culmination	<i>She laughed.</i>  → No culmination

Non-Culminating Perfectives (NCP)

Several languages have telic verbs whose perfective forms do not entail culmination (Altshuler, 2014; Koenig & Chief, 2008; Martin & Demirdache, 2020).

Example from Altshuler (2014) in Russian:

- (1) Ja dočit-yva-l poslednie stročki pis'ma
 I read.up-yva-PST last lines letter
xotja ne do-čita-l ix do konca.
even.though not read.up-PST them until end
“I read the last lines of the letter even though I did not
finish it.”

How do we know NCP are even telic?

How do we know non-culminating perfectives are telic?

- Native speaker-linguist intuition that the event can't go on forever.
- Culmination morphemes (or adverb “completely”). (e.g. Hungarian; Kardos, 2024)
- Implicature of culmination (which is cancellable). (Gyarmathy & Altshuler, 2020)

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Agentivity-Dependent Non-Culmination

In some languages, **agentive** verb phrases allow extra non-culminating readings.

Agentivity involves:

- Subject capable of having desires (often human).
- Usually on purpose.
- Usually involves action.

Agentivity-Dependent Zero-Change Readings

In particular, Demirdache and Martin (2015) observe that a **zero-change** reading often requires agentivity, e.g.:

- (2) a. Yuēhàn shāo le tā-de shu, dàn méi
Yuēhàn burn PFV 3SG-DE book but NEG
shāo-zháo.
burn-touch
“Yuēhàn burned his book, but it didn’t get burned
at all.”
- b. # Huǒ shāo le tā-de shu, dàn méi shāo-zháo.
fire burn PFV 3SG-DE book but NEG burn-touch
Intended: “The fire burned his book, but it didn’t
get burned at all.”

(Though see Nederveen (2025) for a Salish counterexample.)

Why are agentive sentences special?

Martin and Schäfer (2012): Non-culminating perfectives involve modality with universal force, but different flavor:

- Non-agentive: In all worlds compatible with the actual circumstances, the event culminates (includes actual world).
- Agentive: In all worlds where the agent achieves their goal, the event culminates.

→ The agent sets their own conditions for what counts as achieving their goal; event actuality relegated to worlds where they succeed.

Why are agentive sentences special?

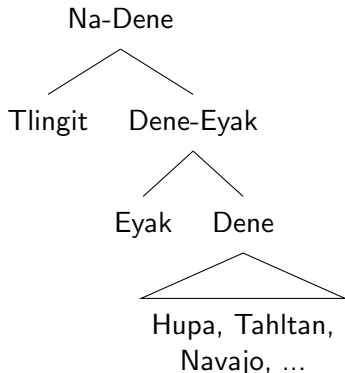
Martin (2015): Non-culminating perfectives always have the same modality regardless of agentivity; however, in the *actual* world the event has to start, and be clearly on-its-way somewhere, and these requirements are met differently:

- Agentive: Event can start before change of state, and it's clearly on-its-way from agent's action.
- Non-agentive: Event starts when change of state starts; also, until then, it's not clear where it's on-its-way to.

→ The agent's actions provide a new way for the event to make partial progress.

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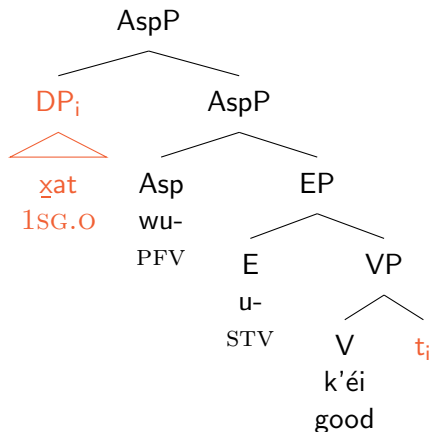
Tlingit (*Lingít* [łín'kít])



- About 100 fluent speakers as of 2018. (Twitchell, 2018)
- Numerous revitalization programs, including language nest, school and university classes, adult classes. (Burge, 2024)

Note on Scope in Tlingit Verbs

Tlingit verb words include syntactic material at least up to aspect (Crippen, 2019), but large non-root elements are moved out, so surface order may not reflect semantic scope.



- (3) xat wookéi
1SG.O PFV.good
“I was good”
(Eggleston, 2017)

Tlingit Perfective and Imperfective

- (4) Loon awliyéx. Laax tsu awliyéx.
bark 3>3.PFV.make red.cedar also 3>3.PFV.make
“He carved bark. He carved red cedar.”
(Dauenhauer & Dauenhauer, 1987)
- (5) Woosh gunayáade át; daa sá yan wulihásh áwé
RECIP different thing; what Q ground PFV.drift FOC
alyéix.
3>3.IPFV.make
“Different kinds of material; whatever had drifted ashore is
what he carved.” (Dauenhauer & Dauenhauer, 1987)

Tlingit Culmination Morpheme

Tlingit perfectives don't have to culminate; culmination is expressed through the **culmination morpheme** **yan=**.

- (6) Yáat'aa x'úx' yeetúw, ch'a aan ku.a tlíl
this.one book PFV.2SG.S.read nonetheless however not
yan yitóow.
CPLV= PFV.2SG.S.read
"You were reading [lit. you read] this book, but you never
finished it." (SJ)

Aside: Perhaps no zero-change reading in Tlingit?

I haven't checked this, but at least this pluractional imperfective form doesn't allow zero-change.

- (6) # Ldakát yá kaa yahaayí áyá dax
all PROX person photo FOC DISTR=
xasagánt', ch'a aan ku.a a xoo
1SG.S.CSV.burn.REP nonetheless however 3N midst
aa tlíl #(k'idéin) dax ugán.
PART NEG #(well) DISTR= IRR-burn
"I'm burning all these pictures, but there are some that
don't burn #(well)." (SJ)

Tlingit Culmination Depends(*) on Telicity

In general *yan*= is only(*) possible on telic verbs.

(7) # **Yan** uwatáa.

CPLV= PFV.sleep

Intended: “She’s done sleeping.”

(SJ)

(*) ...with one exception in the next section.

Tlingit Culmination Depends(*) on Scales

In fact, *yan=* is only(*) possible with verbs that involve **scales of change**.

- (8) A yáx̣ ḳu.aa déix̣ ḳaa x̣'oos uwawát.
it with however two person foot PFV.grow.up
“Then he grew two feet [taller].” (MB, MJ)
- (9) Yeedát/yeedít ḳu.aa yan uwawát.
now however CPLV= PFV.grow.up
“Now he’s all grown up.” (SJ, MB, MJ)

(*) ...with one exception in the next section.

Tlingit Culmination Is a Measure on a Scale

In fact, *yan=* is only(*) possible with verbs that involve **scales of change**, and is incompatible with measures of change besides itself.

- (10) A yáx ku.aa déix kaa x'oos (**yan*)
it with however two person foot (*CPLV=*)
uwawát.
PFV.grow.up
“Then he grew two feet [taller].” (MB, MJ)
- (9) Yeedát/yeedít ku.aa *yan* uwawát.
now however *CPLV=* PFV.grow.up
“Now he’s all grown up.” (SJ, MB, MJ)

Tlingit Culmination as Maximizing on Scales

$$\llbracket \textit{yan} \rrbracket = \llbracket \textit{completely} \rrbracket = \\ \lambda g_{\text{change scale}} \cdot \lambda x_{\text{entity}} \cdot \lambda e_{\text{event}} \cdot g(x)(e) = \max(g)$$

Tlingit Culmination as Maximizing on Scales

$$\llbracket \text{yan} \rrbracket = \llbracket \text{completely} \rrbracket = \\ \lambda g_{\text{change scale}} \cdot \lambda x_{\text{entity}} \cdot \lambda e_{\text{event}} \cdot g(x)(e) = \max(g)$$

With a degree achievement (à la Kennedy & Levin, 2008)

like *wat* “grow up”:

$$\llbracket \text{wat} \rrbracket = \\ \lambda x_{\text{entity}} \cdot \lambda e_{\text{event}} \cdot \text{the degree to which } x \text{ matures over the course of } e$$

Tlingit Culmination as Maximizing on Scales

$$\llbracket \text{yan} \rrbracket = \llbracket \text{completely} \rrbracket = \\ \lambda g_{\text{change scale}}. \lambda x_{\text{entity}}. \lambda e_{\text{event}}. g(x)(e) = \max(g)$$

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$$\llbracket \text{wat} \rrbracket = \\ \lambda x_{\text{entity}}. \lambda e_{\text{event}}. \text{the degree to which } x \text{ matures over the course of } e$$

Then:

$$\llbracket \text{yan wat} \rrbracket = \\ \lambda x_{\text{entity}}. \lambda e_{\text{event}}. \text{the degree to which } x \text{ matures over the course of } e \\ \text{is the maximum of that scale.}$$

Tlingit Culmination as Maximizing on Scales

$$\llbracket \text{yan} \rrbracket = \llbracket \text{completely} \rrbracket = \\ \lambda g_{\text{change scale}} \cdot \lambda x_{\text{entity}} \cdot \lambda e_{\text{event}} \cdot g(x)(e) = \max(g)$$

With a degree achievement (à la Kennedy & Levin, 2008)

like *wat* “grow up”:

$$\llbracket \text{wat} \rrbracket = \\ \lambda x_{\text{entity}} \cdot \lambda e_{\text{event}} \cdot \text{the degree to which } x \text{ matures over the course of } e$$

Then:

$$\llbracket \text{yan wat} \rrbracket = \\ \lambda x_{\text{entity}} \cdot \lambda e_{\text{event}} \cdot \text{the degree to which } x \text{ matures over the course of } e \\ \text{is the maximum of that scale.}$$

Add perfective aspect to get *yan uwawát* “He’s all grown up”.

This echoes Koenig and Chief’s (2008) account of Mandarin NCP.

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Agentive atelic verbs can “culminate” in Tlingit

- (11) **Yan** xwaditúw.
CPLV= PFV.1SG.S.APSV.read
 “I’m finished reading.” (SJ)

Atelic verb “culmination” requires agent

Context: *The speaker has a machine that knits automatically. Without anyone helping it, it can knit blankets. They are watching it work, but suddenly it stops working.*

- (12) # Tléik'! **Yan** kamdzinéi!
no **CPLV=** PFV.APSV.knit
Intended: “Oh no! It quit knitting!” (MB, MJ)
Not shown: This is fine with the verb for “to quit.”

Atelic verb “culmination” requires agent to stop trying

Context: *A little girl is talking about her baby brother. He was punching the table, but their mom picked him up and took him away.*

- (13) # Wé nadáakw **yan** aawagwál, ch'a aan
that table **CPLV=** 3>3.PFV.hit, nonetheless
akoo.aakw.
3>3.try
“He quit punching the table, but he’s trying again.”
(MB, MJ)
Not shown: This is fine with the verb for “to quit.”

Atelic verb “culmination” has to be in agent’s own opinion

Context: *A child is popping balloons from a party. The child's father tells him to stop.*

- (14) # Aan **yan** keensi.úk'.
with.it CPLV= 2SG.S.PFV.APPL.play
Intended: "You're done playing with them."
(SJ, MB, MJ)
- (15) Aan **yan** keensi.úk' gi?
with.it CPLV= 2SG.S.PFV.APPL.play Q
"Are you done playing with them?"
(SJ, MB, MJ)

Telic VPs can have either kind of “culmination”

Context: *A woman is going to the library, and her brother asks her to return a book for her. She points out that he's only read half of it, but he says he doesn't like it and wants to return it. This is the brother speaking.*

- (16) % Wé x'úx' **yan** x̣waatúw. **Tleil** aḵ tuwáa
that book **CPLV=** PFV.1SG.S.read **not** my desire
ushgú **yan** x̣watóowu, wé
IRR.CSV.happy **CPLV=** PFV.1SG.S.read.C that
sh kalneegí tléil aḵ tuwáa ushgú.
story not my desire IRR.CSV.happy
“I'm done reading that book. I don't want to finish it
because I didn't like the story.” (MB, MJ; # SJ)

Tlingit Culmination vs. Demirdache and Martin (2015)

	D&M 2015 Perfectives	Tlingit Perfective	Tlingit Perfective with <i>yan</i>
Zero Change (Non-Agentive)	×	×	×
Zero Change (Agentive)	✓	×	×
Partial Change (Non-Agentive)	✓	✓	×
Partial Change (Agentive)	✓	✓	✓

Tlingit Culmination vs. Demirdache and Martin (2015)

	D&M 2015 Perfectives	Tlingit Perfective	Tlingit Perfective with <i>yan</i>
Zero Change (Non-Agentive)	✗	✗ (?)	✗
Zero Change (Agentive)	✓	✗ (?)	✗ (?)
Partial Change (Non-Agentive)	✓	✓	✗
Partial Change (Agentive)	✓	✓	✓

Combining Agentivity and Scale-Based Culmination (A)

If Tlingit agentive perfectives *do* allow zero-change readings:
Agentivity (perhaps agentive v) adds another scale of agent participation, with no relation to the original scale.

$\llbracket v \rrbracket =$
 $\lambda P_{\text{predicate}}. \lambda x_{\text{entity}}. \lambda e_{\text{event}}.$ the proportion of actions taken, out of the actions x wants to take which (in normal circumstances) would make $F(e)$ happen.

Combining Agentivity and Scale-Based Culmination (A)

$\llbracket v \rrbracket =$
 $\lambda P_{\text{predicate}} \cdot \lambda x_{\text{entity}} \cdot \lambda e_{\text{event}} \cdot$ the proportion of actions taken, out of the actions x wants to take which (in normal circumstances) would make $F(e)$ happen.

Positive progress along *this* scale is enough for normal perfective meaning (with zero change on the original scale), e.g. *Yuē hàn shāo le tā-de shu* “Yuē hàn burned his book”:

$\llbracket [Yuēhàn [v shāo tā-de shu]] \rrbracket =$
 λe . Yuēhàn takes some actions he wants to take which would normally make burning his book happen.

Combining Agentivity and Scale-Based Culmination (A)

$\llbracket v \rrbracket =$

$\lambda P_{\text{predicate}} \cdot \lambda x_{\text{entity}} \cdot \lambda e_{\text{event}} \cdot$ the proportion of actions taken, out of the actions x wants to take which (in normal circumstances) would make $F(e)$ happen.

Culmination on this scale is enough for a *yan* perfective, e.g. *yan xwaditúw* “I’m done reading”:

$\llbracket [\underline{x} \text{ } [\textit{yan} [v \textit{ ditóow}]]] \rrbracket =$

λe . I take all the actions I want to take which would normally make me-reading happen.

Since *yan* can modify non-scalar agentive verbs, something like this is necessary anyway. This also derives its ambiguity with scalar agentive verbs.

Combining Agentivity and Scale-Based Culmination (B)

If Tlingit agentive perfectives *don't* allow zero-change readings, we still need agent-determined culmination, but progress toward this culmination entails progress along the original scale.

In this case, Tlingit is genuinely different from languages in Demirdache and Martin (2015), and I don't have a satisfying story for their similarities.

I can modify the previous story to get the entailment to work.

$\llbracket v \rrbracket =$
 $\lambda P_{\text{predicate}} \cdot \lambda x_{\text{entity}} \cdot \lambda e_{\text{event}} \cdot$ the proportion of actions taken, out of the actions x wants to take which (in normal circumstances) would make $F(e)$ happen; 0 if $F(e)$ does not happen.

Combining Agentivity and Scale-Based Culmination (C)

Third possibility: Tlingit agentive perfectives allow zero-change readings, but with *yan* they require positive change. We still have a separate agent scale, but now *getting to the end of that scale entails starting the original event*.

$\llbracket v \rrbracket =$
 $\lambda P_{\text{predicate}} \cdot \lambda x_{\text{entity}} \cdot \lambda e_{\text{event}} \cdot$ The proportion of x 's intended actions during e that they take, where doing everything they intend would make $F(e)$ happen.

E.g.:

$\llbracket [Yu\bar{e} \text{ hàn } [v \text{ shāo } t\bar{a}-de \text{ shu }]] \rrbracket =$
 λe . Yuē hàn takes some actions e toward what he intends, (where if he did them all he would burn his book).

$\llbracket [\underline{x} \text{ } [yan \text{ } [v \text{ ditóow }]]] \rrbracket =$
 λe . I took all my intended actions during e , where doing them all would make my book-reading happen. ($\rightarrow$ So it happened.)

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Summary

- In several languages, agentivity facilitates a zero-change reading of non-culminating perfectives.
- In Tlingit, agentivity allows you to use a scale-based culmination morpheme when the end of the usual scale isn't reached.
- This makes sense if agents change the scale on which event progress is measured.

Summary

- In several languages, agentivity facilitates a zero-change reading of non-culminating perfectives.
- In Tlingit, agentivity allows you to use a scale-based culmination morpheme when the end of the usual scale isn't reached.
- This makes sense if agents change the scale on which event progress is measured.

Predictions:

- Changing the agent should allow a second “culmination”.
- For non-scalar base verbs, comparing “who did X more” should be about progress on agent-participation scale.
- If Tlingit agentive perfectives allow zero-change readings, so should ones with *yan*, and also, agentive *progressives*.

Acknowledgments

Aatlein gunalchéesh to *Anyàłashâsh* Sam Johnston, *Shuwuteen* Margaret Bob, and *Kudagaan* Madeleine Jackson for their language expertise.

To James Crippen, Seth Cable, Luis Alonso-Ovalle, the McGill Syn-Sem Reading Group, the Na-Dene Reading/Working Group, and the audience at CLA for insightful comments.

To Fiona Smith for help with corpus data.

To Millie Hall, Tim Hall, Andy Hall, Corinne Johnston, and Connie Jules for huge assistance during fieldwork trips.

To the Fonds de Recherche du Québec, the American Philosophical Society, and the Jacobs Research Fund for financial support.



Fonds de recherche – Nature et technologies
Fonds de recherche – Santé
Fonds de recherche – Société et culture

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